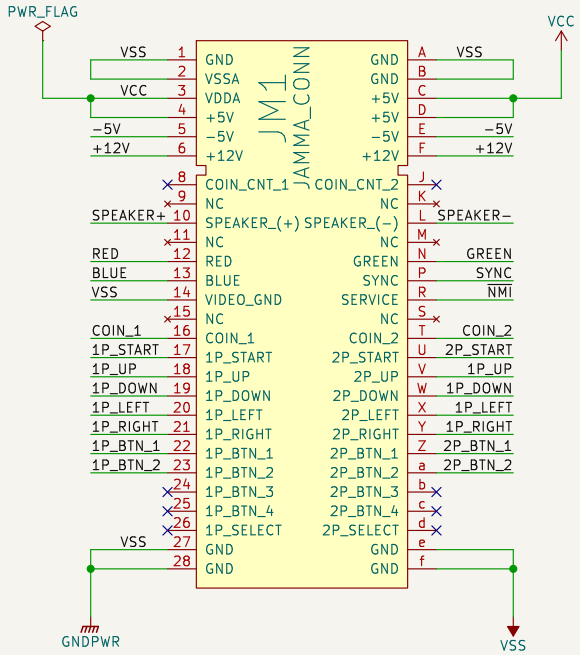
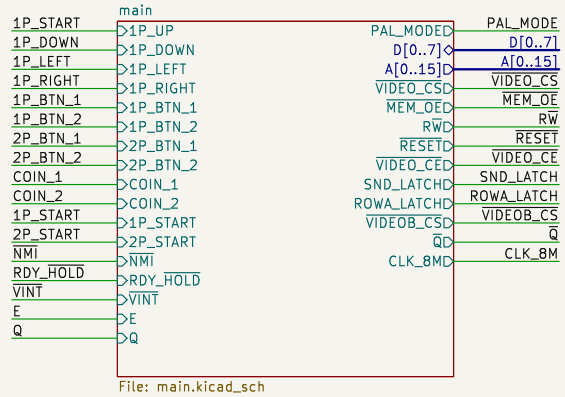
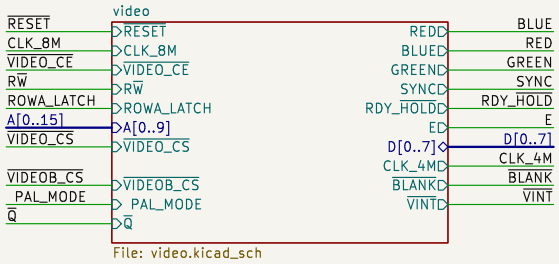
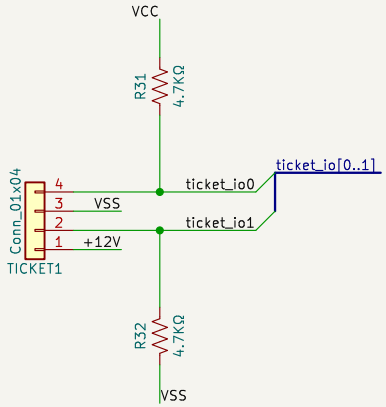
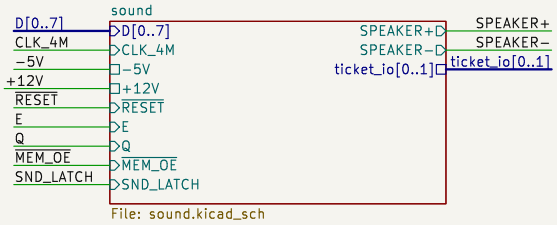










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Sheet: /
File: bowl.kicad_sch

Title: Beer Frame Bowling

Size: A4	Date: 2022-06-06	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 1/5

VC1
TMS34061FNL
VIDEO CONTROLLER

HAND REPAIRED
PIN 5 TO R_{W
PIN) 6 TO PL (U27)

NOT POPULATED

U20E 74LS86

74LS76 U3C

74LS10 U4D

JOTEGO
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Sheet: /video/
File: video.kicad_sch

Title: Beer Frame Bowling

Size: A3	Date: 2022-06-06	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 2/5

SND_D[0..7]

D[0..7]

U33 74HCT374

U29 MC68B09E

U32 74LS138 address decoder

SNDA[0..15]

U31 HM6116P-3 2kB RAM

U30 27C256

U89 74LS04

U2B 74LS74

R2 4.7KΩ

U39 DAC0832LCN

YM34

YM2203

YM35 YM3014

ticket_io[0..1]

CLK_4M

RESET

SND_FIRQ

SND_LATCH

LATCH_RD

IRQ

RW

RD

ROM_CS

RAM_CS

FM_CS

PCM_WR

R28 390Ω

CR2 LED_Small_ALT

C24 EC0011

C1 CP1

R18 1KΩ

R17 1KΩ

R16 1KΩ

R22 150KΩ

R21 150KΩ

R20 150KΩ

R19 150KΩ

R30 36KΩ

636KΩ

U36A U36B U36C U36D U37A U37B U37C U37D MC3403N

+12V

−5V

C23 EC0011

U37E MC3403N

U38 MB3730

C8 H1_8810

R27 1KΩ

C9 4.7µF

C12 CP1

C11 22µF

RV3 R_POT_US

R25 1KΩ

R24 75Ω

C13 1000uF

C2 C4 C17 C18 C19 C20 C21 C22 EC0011

C16 EC0011

C14 EC0011

C15 EC0011

SPEAKER−

SPEAKER+

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Sheet: /sound/

File: sound.kicad_sch

Title: Beer Frame Bowling

Size: A3 | Date: 2022-06-06 | **Rev: José Tejada G.**

KiCad E.D.A. 8.0.9 | Id: 3/5

U5 8MHz CLK_8M E/B OUT GND V+ TD308C/TD1100C VSS VCC

U3A U3B 74LS76 U2A 74LS74 RDY_HOLD R39 4.7KΩ Q Q̄ E E

U11 74HCT374 D0 D1 D2 D3 D4 D5 D6 D7 BANK_CS VSS VCC ROM_A14 ROM_A15 VIDEO_CS PAL_MODE U8B 74LS04 VIDEOB_CS

U18 address decoder 74LS138 U26C 74LS32 A14 A15 Q E U8A 74LS04 MEM_OE CEA CEB CEC CED CEE CEF CEG CEH CEI

D[0..7]

U21 74LS191 U23 74LS244 R4 4.7KΩ R5 4.7KΩ R6 4.7KΩ R7 4.7KΩ 2P_BTN_1 2P_BTN_2 2P_START COIN_2 PL VSS

U25A 74LS74 U17C 74LS14 1P_UP C26 EC0011 R42 1KΩ

U17E 74LS14 1P_LEFT C28 EC0011 R44 1KΩ

U22 74LS191 U24 74LS244 R8 4.7KΩ R9 4.7KΩ R10 4.7KΩ R11 4.7KΩ 1P_BTN_1 1P_BTN_2 1P_START COIN_1 IN1

U25B 74LS74 U17D 74LS14 1P_DOWN C25 EC0011 R41 1KΩ

U17F 74LS14 1P_RIGHT C27 EC0011 R43 1KΩ

U1 MC68B09E R15 4.7KΩ RESET NMI IRQ FIRQ VINT NMIC HALT E Q LIC AVMA R/W TSC BA BS BUSY RW A[0..15]

U9 address decoder 74LS138 A11 A13 ROM_A143 A14 A15 BANK_CS GROM_CS VIDEO_CE SND_LATCH ROWA_LATCH PL IN0 IN1 U17A 74LS14 CEI

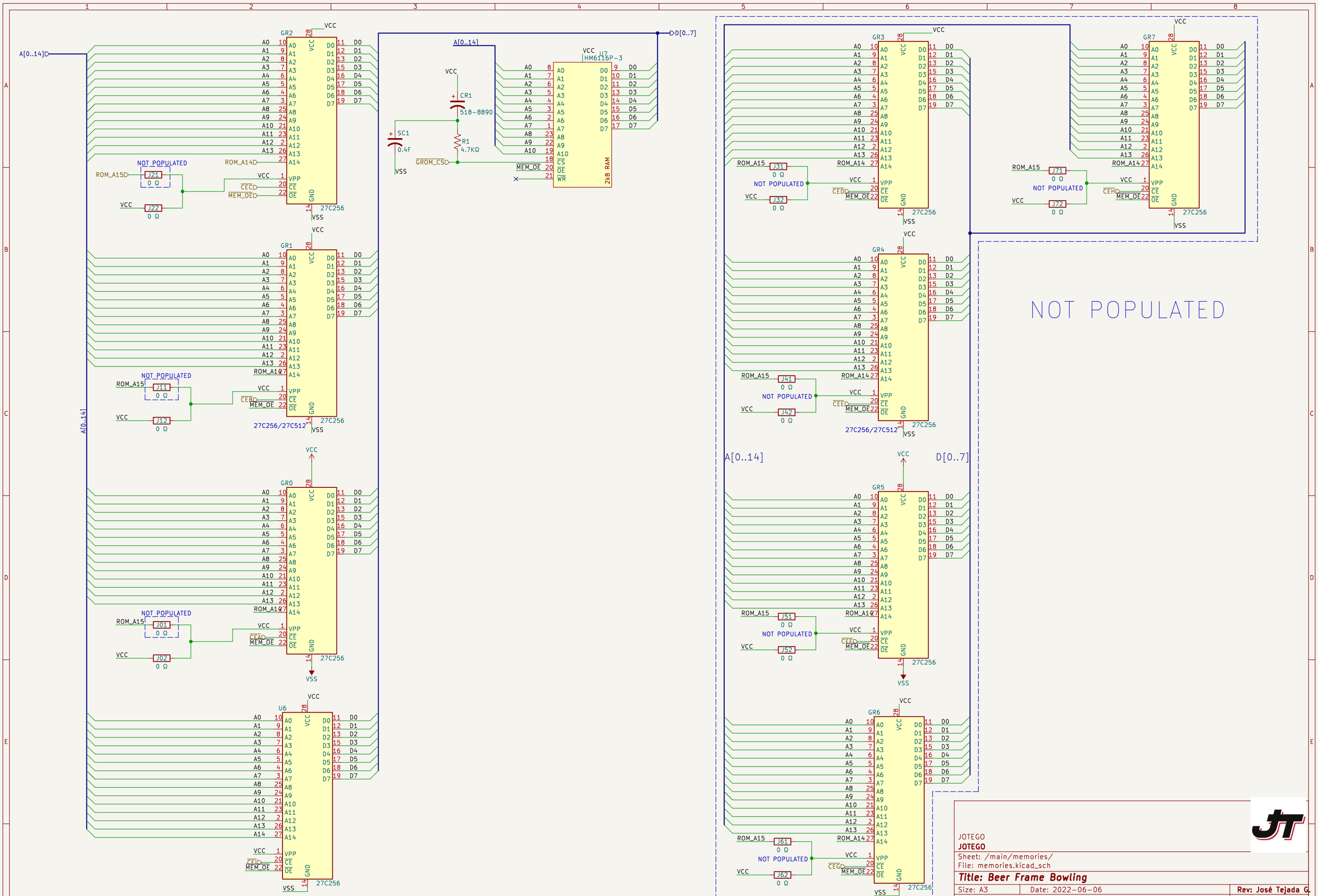
U27 74LS191 R40 4.7KΩ CONNECTION REPAIRED TO_U4_PIN4_ RESET

U28 NE555 R13 10KΩ R14 10KΩ C5 EC0011 C6 EC0011

memories.kicad_sch

U25C 74LS74 U26E 74LS32 U8G 74LS04 U2C 74LS74 U17G 74LS14

JOTEGO
JOTEGO
Sheet: /main/
File: main.kicad_sch
Title: Beer Frame Bowling
Size: A3 | Date: 2022-06-06 | **Rev: José Tejada G.**
KiCad E.D.A. 8.0.9 | Id: 4/5



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Sheet: /main/memories/

File: memories.kicad_sch

Title: Beer Frame Bowling

Size: A3	Date: 2022-06-06	**Rev: José Tejada G.**
KiCad E.D.A. 8.0.9		Id: 5/5